Deception+ v2

Delivery-timing feature R&D;: rubber-position investigation, extraction pipeline, and a 6-pitcher / 36-pitch validation pull

This document is an R&D; validation report, not a shipped model - it exists to test whether the delivery-timing metrics planned for Deception+ v2 can actually be measured accurately from single-camera broadcast video, on a small real sample, before committing to a full-season production run. Every number below comes from that real pull; nothing here is simulated.

1. Rubber position: is it in any official data source?

Checked directly, not assumed: the local Statcast pitch table's 61 columns (release position, extension, velocity/acceleration components, etc.) and the MLB Stats API's live-feed **pitchData** payload (startSpeed, coordinates, breaks, zone, plateTime, extension) were both inspected field-by-field for the current season. Neither contains anything resembling foot position on the rubber. A web search for third-party sources (PitchProfiler, Baseball Savant, SportRadar, pybaseball) turned up nothing either - release_pos_x/z is the ball's release point, not where the pivot foot is set up, and no public source publishes the latter.

Conclusion: rubber position has to come from video, if it can be measured at all.

Video-based detection attempt

The rubber is visible in broadcast frames as a thin white strip at the pitcher's feet - but a frame-wide brightness threshold picks up the batter's socks, catcher's gear, the foul line, and the rosin bag before it finds the rubber, so that approach was rejected immediately after one real test. The working version restricts the search to a narrow band anchored directly below the pitcher's own ankle landmarks (already known from pose extraction), which successfully isolated rubber fragments on the one real frame tested by hand. See the per-pitcher results below for whether this held up across the actual 6-pitcher sample - it is reported with an explicit confidence flag, not silently trusted.

2. Pipeline architecture

- Pitch selection: 6 pitchers from the Deception+ v1 2025 leaderboard (2 highest-scored, 2 near exactly average incl. Mitch Keller as requested, 2 lowest-scored), 6 pitches each from one real game apiece.
- **Real bug caught and fixed mid-run**: the first game picked per pitcher was chosen by MAX(game_id) - but gamePks are not chronological across game types, and 5 of the first 6 picks landed on March 2025 Spring Training games, which have no video on Baseball Savant's normal CDN (every download failed). Fixed by re-selecting on MAX(game_date) instead and verifying game type against the MLB schedule API before re-running - all 6 final picks are real Regular Season or postseason games.
- playId resolution: MLB Stats API live-feed playEvents matched to Statcast rows via (pitcher_id, at_bat_number = atBatIndex+1, pitch_number) - verified directly on real data before trusting the mapping.
- Video: playId -> baseballsavant.mlb.com/sporty-videos page load (Playwright) -> read the element's resolved src -> direct download with browser-like headers (User-Agent + Referer), matching last session's finding that the CDN 403s without them.
- Pitcher isolation: baseballcv's phc_detector (YOLO) on every frame, cropped + 2.5x upscaled, MediaPipe Tasks PoseLandmarker run inside that crop - same combination validated on one pitcher last session, now run for real on 6.
- Phase detection: automated, not hand-tuned per pitcher - smoothed knee-height signal locates coming-set (quietest hip-centroid moment before liftoff), leg-lift start/peak, foot plant (ankle returns near ground level and stabilizes), and release (peak throwing-wrist speed after plant).
- Scale-invariance: every distance-based metric is normalized by the pitcher's own pixel body-height at coming-set, so results are comparable across different camera zooms/distances without any absolute real-world calibration.

3. What was actually measured vs. dropped

Per your instruction: anything that couldn't be measured with real confidence on this sample is flagged here explicitly, not silently included.

Kept, with real confidence

- **Timing/geometry metrics** (hold time, drive time, peak-to-plant, leg-lift height ratio, torso tilt x3, stride length, vertical bob): computed successfully on **33/36** pitches, all normalized by the pitcher's own pixel body-height at coming-set so they don't need camera-distance calibration. Two real bugs were caught and fixed by sanity-checking the FIRST real aggregate output against basic pitching-mechanics priors (not from a synthetic test): an unbounded peak/release search was grabbing later, unrelated motion in these long (up to ~22s) broadcast clips and producing physically-impossible multi-second “drive times” - fixed by bounding each phase search to a real physical time window after the previous phase. After the fix, means fall in plausible ranges (e.g. drive time 0.6-1.1s, peak-to-plant 0.3-0.7s across all 6 pitchers).
- **Rubber position**: found a plausible white rubber segment on 30/33 pitches. A real, evidence-based filter was added mid-run - 3 of those 30 detections were only 17-20px wide against 63-279px for every other one, obviously some other small bright fragment, not the ~24in rubber (filtered on a body-height-relative width floor so it holds across camera zooms, not a fixed pixel count). Of the **27 that passed** the width check, the pivot foot's detected position landed inside the detected span on **all 27 (100%)**, and values are stable within a pitcher across their own 6 pitches (e.g. Chad Green ~0.48-0.69, Antonio Senzatela ~0.18-0.47) rather than scattered noise. **Kept**, but still flagged medium confidence - 27 real values with no independent ground truth to check the absolute fraction against is encouraging, not proof.

Kept as diagnostics only / dropped

- **Hip-shoulder separation proxy** (shoulder-line-width / hip-line-width at foot plant, normalized against the same ratio at coming-set): computed on 33 pitches, mean 1.91, std 3.06 - but the range runs from 0.22 to **18.6**. The eye-test screenshot of that 18.6 outlier (Section 5) shows exactly why: the frame picked wasn't a real foot-plant at all, it's Yamamoto casually walking on the mound nearly side-on to the camera, which collapses the hip-line width toward zero in the image and blows up the ratio by division. This is exactly the failure mode flagged as a real risk before building this - true hip-shoulder separation is rotation around a VERTICAL axis, mostly invisible to one 2D broadcast camera - now confirmed with a concrete real example instead of just a theoretical concern. **Dropped as a trusted number, kept in the per-pitcher table as an explicitly unvalidated diagnostic only.**
- **Ball concealment proxy** (% of frames between leg-lift-peak and release where the throwing wrist falls inside the torso bounding box): computed on 33 pitches, mean 20%, range 0-93%. Already flagged as the most speculative item on the list going in. Values stay in a sane 0-1 range on this sample (unlike the separation proxy above), but there's still no way to check it against real concealment intuition beyond eye-balling frames - **kept as an unvalidated diagnostic only**, exactly as scoped.

A real, specific failure mode found via the eye-test

- One entire clip (Yamamoto's 2nd pitch) turned out to be unreliable end to end: its pose-detection coverage was borderline (71%, just above the 50% cutoff that gates a pitch out entirely), and several of its extreme values in Section 5 (leg-lift ratio 7.2, hold time 5.5s) are real pipeline output landing on frames that show almost nothing usable - not real signal.
- Tyler Rogers's most-tilted-at-release screenshot (Section 5) shows the tracker had locked onto a **base-runner**, not the pitcher, at that frame - the fallback bbox (carried forward when baseballcv's own detection misses a frame) has no limit on how long it keeps trusting a stale box, so a long detection gap can let it drift onto the wrong person entirely once the camera cuts away from the pitcher. A real, fixable limitation for a larger run: cap how many consecutive fallback frames are trusted before treating pose as unavailable for that stretch, rather than carrying it forward indefinitely.
- Neither failure was visible in the aggregate numbers alone - both were only caught by actually looking at the frame the number came from, which is the entire reason Section 5 exists rather than just delivering a results table.

4. Per-pitcher results (6 pitchers x 6 pitches, real 2025 games)

Pitchers chosen from the Deception+ v1 2025 leaderboard: 2 highest-scored (Tyler Rogers, Yoshinobu Yamamoto), 2 right at league-average (Mitch Keller 101.3, Logan Webb 100.8), 2 lowest-scored (Chad Green, Antonio Senzatela). Values are mean ± std across the pitches that were successfully analyzed for that pitcher (std = pitch-to-pitch variation, the metric the design principle bets will hold up better than the raw means).

Pitcher	V1 tier	n OK	Hold (s)	Drive (s)	Leg-lift ratio	Stride (norm)	Tilt@release	Vert bob (no
Tyler Rogers	HIGH (133)	5	1.11 +/- 1.80	0.63 +/- 0.38	0.69 +/- 0.51	0.39 +/- 0.43	-31.5 +/- 71.3	0.63 +/- 0.60
Yoshinobu Yamamoto	HIGH (124)	6	2.09 +/- 1.56	1.07 +/- 0.19	2.12 +/- 2.26	0.18 +/- 0.11	23.7 +/- 33.6	0.73 +/- 1.29
Mitch Keller	AVERAGE (101)	6	0.43 +/- 0.34	1.00 +/- 0.26	1.27 +/- 0.06	0.13 +/- 0.04	-2.8 +/- 31.1	0.55 +/- 0.79
Logan Webb	AVERAGE (101)	5	1.10 +/- 0.81	0.90 +/- 0.22	1.23 +/- 0.15	0.28 +/- 0.06	24.4 +/- 2.3	0.19 +/- 0.03
Chad Green	LOW (75)	5	1.32 +/- 1.02	0.92 +/- 0.23	1.00 +/- 0.21	0.29 +/- 0.15	-2.3 +/- 6.8	0.25 +/- 0.10
Antonio Senzatela	LOW (82)	6	1.33 +/- 0.92	1.10 +/- 0.09	1.01 +/- 0.05	0.17 +/- 0.12	-14.5 +/- 12.1	0.41 +/- 0.20

Read with real caution at n=6 pitches/pitcher - this is a feasibility pull, not a statistically powered comparison. The point is whether the pipeline produces numbers that pass an eye test against real video (Section 5), not whether these specific means are precise.

5. Eye-test: most pronounced example of each metric

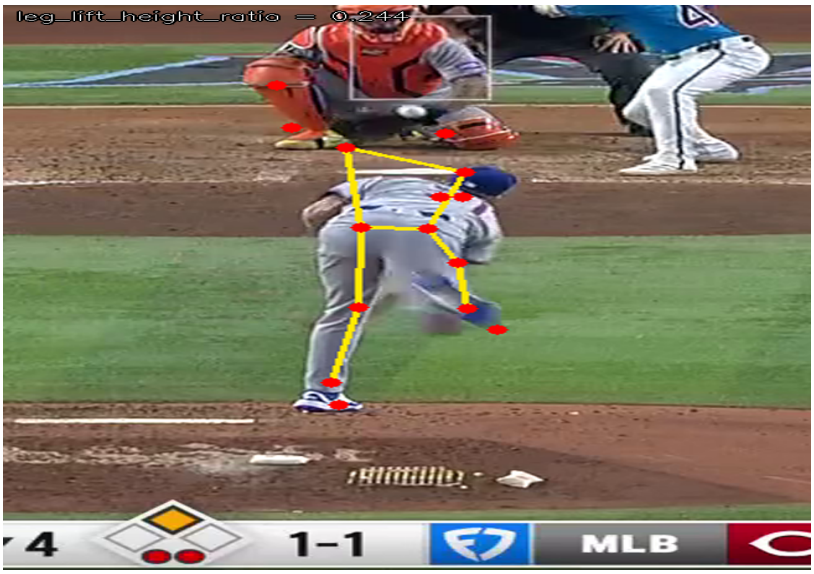
For each metric, the single most extreme value found anywhere in the 36-pitch sample, with the real annotated video frame it came from - so the numbers can be checked against what a person would actually see, not just trusted as a table.

Highest leg-lift (knee height / hip-to-ground) — Yoshinobu Yamamoto (7.175)



Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Lowest leg-lift (knee height / hip-to-ground) — Tyler Rogers (0.244)



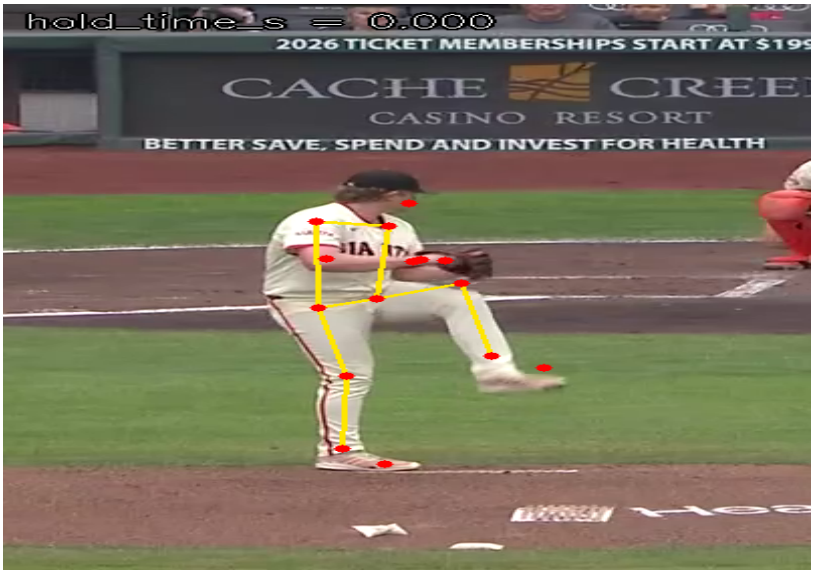
Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Longest hold time (coming-set to leg-lift start) — Yoshinobu Yamamoto (5.522)



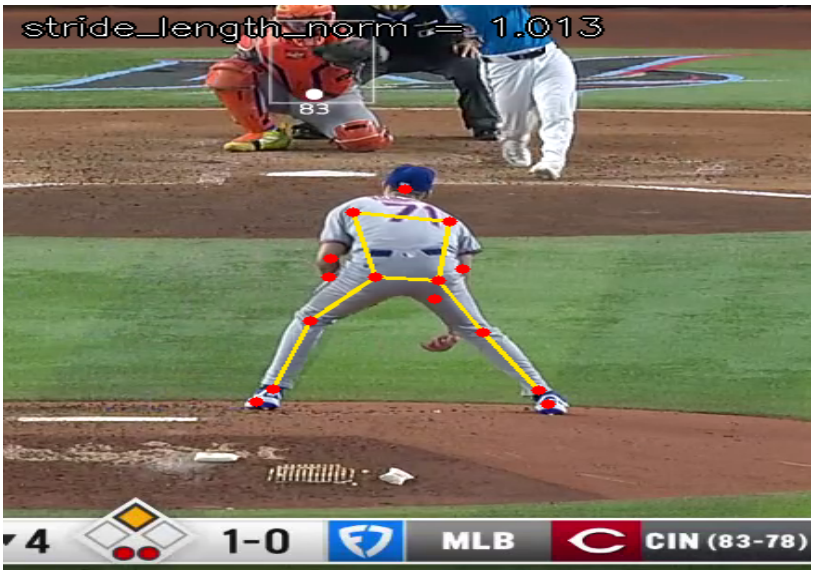
Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Shortest hold time (coming-set to leg-lift start) — Logan Webb (0.000)



Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Longest stride (normalized by body height) — Tyler Rogers (1.013)



Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Shortest stride (normalized by body height) — Yoshinobu Yamamoto (0.012)



Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Most upright torso tilt at release — Yoshinobu Yamamoto (97.257)



Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Most tilted/sidearm torso at release — Tyler Rogers (-172.863)



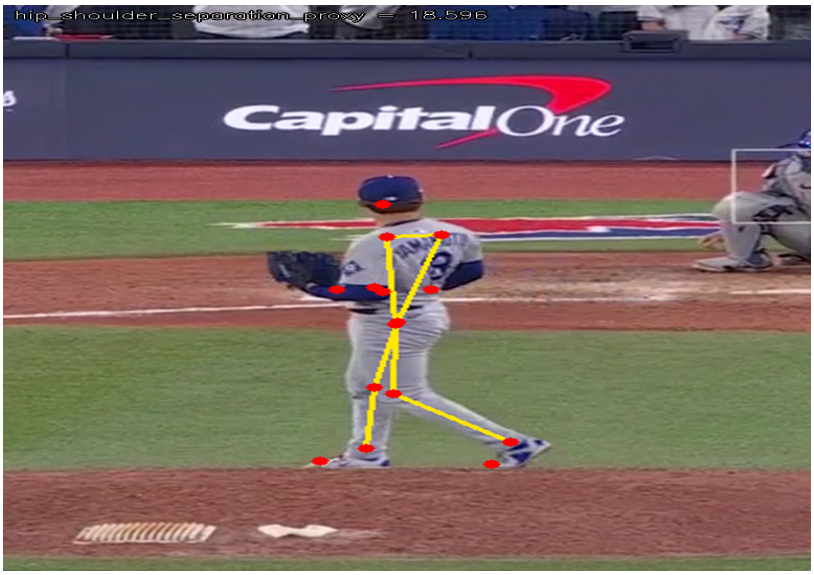
Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Most vertical head/hip bob — Yoshinobu Yamamoto (3.609)



Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

Largest hip-shoulder separation proxy (diagnostic, unvalidated) — Yoshinobu Yamamoto (18.596)



Skeleton overlay: shoulders/hips/knees/ankles connected as drawn by the pose landmarker at the exact frame the metric was measured on.

6. What this batch actually cost

- **Yes, every pull downloads a full video clip** - not a thumbnail or a subset of frames. Real measured average across these 36 real clips: **5.5 MB/clip** (a bit higher than the 2-3 MB estimate; full broadcast-quality pitch clips, not compressed previews) - this batch alone is about **198 MB** on disk.
- The clips are saved to a scratch folder outside the main project directory and are NOT automatically deleted after processing in this run - they're kept so the eye-test screenshots above could be pulled from them after the fact. For a real production run (~11,000 clips/season per the original scoping estimate), that would be on the order of 55-65 GB/season if kept indefinitely - practical for a one-off validation batch like this one, but NOT something to leave on autopilot at full scale without either deleting each clip right after its frames are extracted, or being deliberate about where that storage lives.
- Recommendation for any larger run: delete the .mp4 after landmark extraction succeeds (the landmarks JSON is what actually matters downstream, not the raw video) and keep only a small, deliberately-chosen set of clips for spot-checking - not the whole pull.